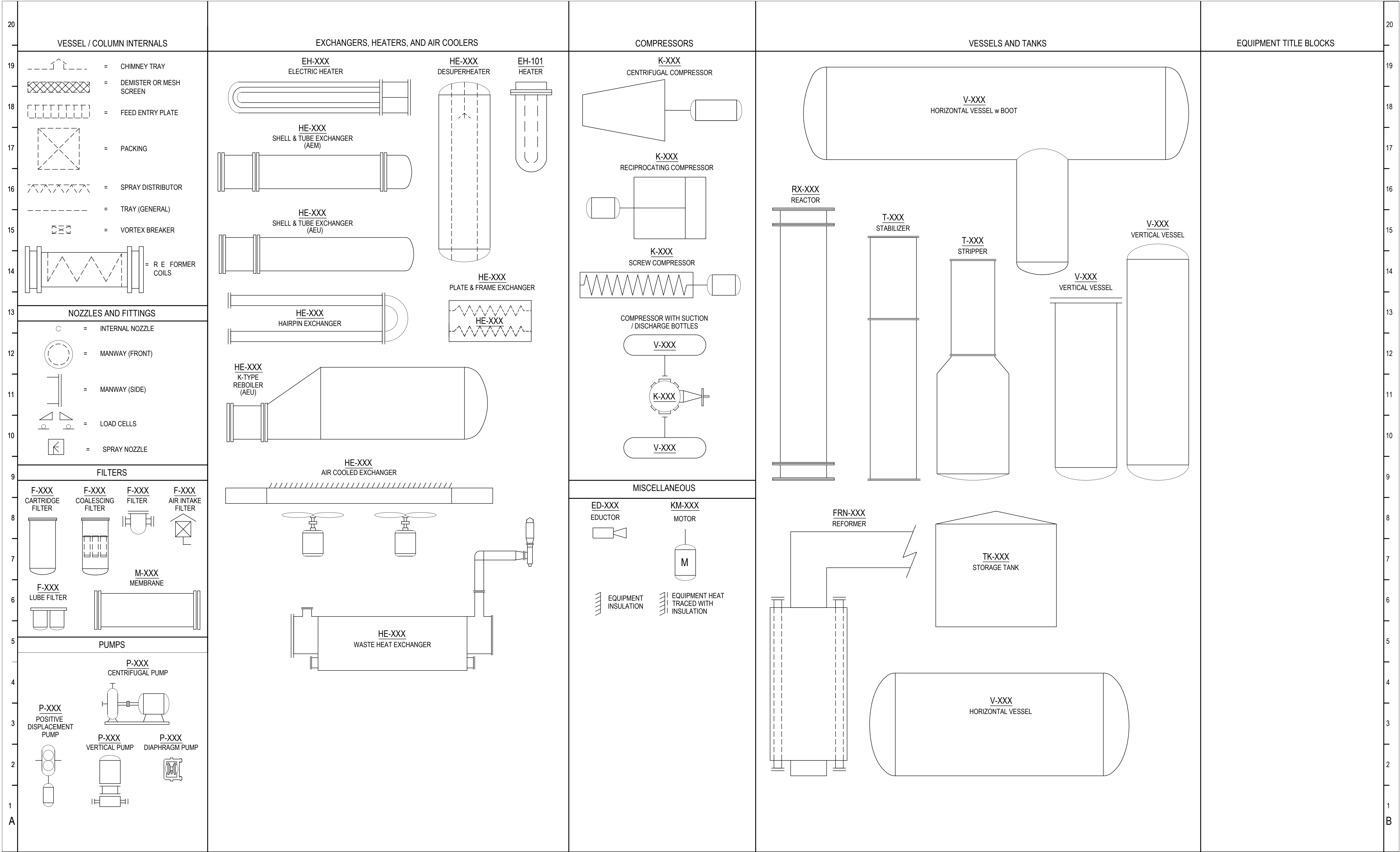


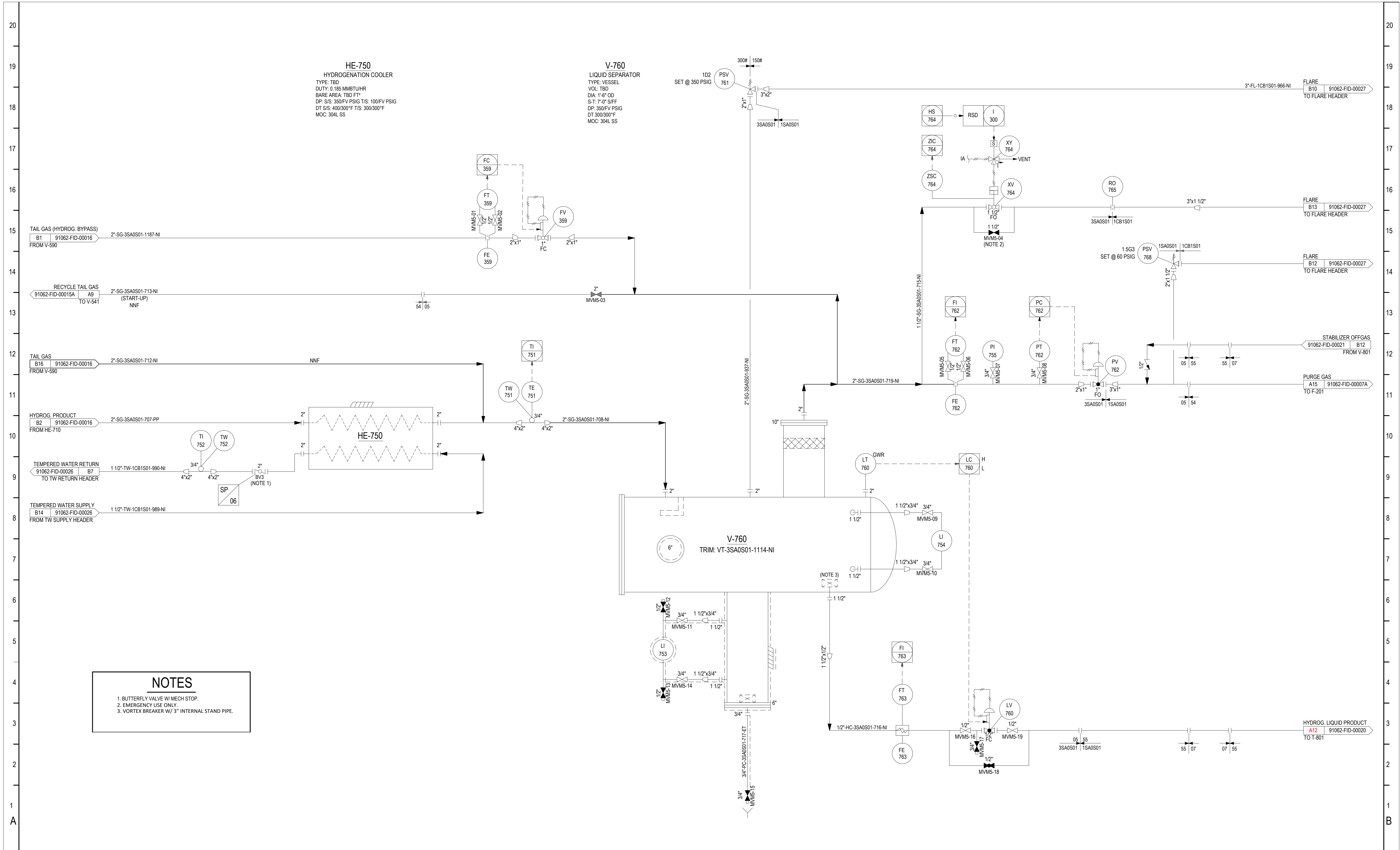
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20	INSTRUMENT FUNCTION LETTERS			ABBREVIATIONS (GENERAL)						INSTRUMENT IDENTIFICATION			GENERAL NOTES			20												
19	(USED INSIDE INSTRUMENT BUBBLES)			AG ABOVE GROUND			ISBL INSIDE BATTERY LIMITS			INSTRUMENT LINE SYMBOLS			1. ALL PIPING VENTS ARE 3/4", EXCEPT AS NOTED.			19												
	SELECT FIRST LETTER FROM TABLE AT LEFT			ATM ATMOSPHERE			LC LOCK CLOSE			CONNECTION TO PROCESS MECHANICAL LINK OR INSTR. SUPPLY INSTRUMENT AIR SIGNAL ELECTRICAL SIGNAL DATA HIWAY / SOFT LINK CAPILLARY TUBING INFRARED SIGNAL INSTRUMENT ARROW			2. ALL PIPING DRAINS ARE 3/4" EXCEPT AS NOTED.			18												
18	A ANALYSIS			BD BLOWDOWN			LFSD LOW-FIRE SHUTDOWN									3. CONTROL VALVE ACTION ON AIR OR ELECTRICITY FAILURE: FO-OPENS, FC-CLOSES, FLC- LOCKS STATIC (AND WOULD FINALLY DRIFT TO CLOSE ON LEAKAGE OF AIR FROM ACTUATOR).												
	B BURNER/COMBUSTION			BF BLIND FLANGE			LO LOCK OPEN						4. ALL RV'S AND VENTS DISCHARGING TO ATMOSPHERE SHOULD HAVE 3MM WEEP HOLE AT LOW POINTS. WEEP HOLE TO BE DIRECTED AWAY FROM AREAS WHERE PERSONNEL MAY BE PRESENT.			18												
	C CONDUCTIVITY (ELECT.)			B.F. BACK FLUSH			LP LOW PRESSURE						5. UNLESS OTHERWISE NOTED, LETTER SYMBOLS FOR INSTRUMENTATION CONFORMS TO I.S.A. ENGINEERING STANDARD S5.1			17												
17	E VOLTAGE (EMF)			BL BATTERY LIMITS			MIN. MINIMUM DISTANCE						6. DEFINITIONS: GRAVITY FLOW - ELEVATIONS DOWNSTREAM NEVER EXCEED INLET ELEVATIONS. LINE MAY CONTAIN POCKETS SLOPE - ELEVATION CHANGES ARE DOWNWARD ONLY. NO POCKETS ARE PERMITTED. SPECIFIC SLOPE REQUIRED ARE SHOWN BY SYMBOL WHERE X HAS UNITS THE SAME AS I NO POCKETS - NO LIQUID POCKETS IN THE LINE.															
	F FLOW RATE			BSD BURNER SHUTDOWN			MW MANWAY						7. LOCATION OF RELIEF VALVES SHALL BE:			17												
	FQ FLOW QUANTITY			CB CONTINUOUS BLOWDOWN			MWP MAXIMUM WORKING PRESSURE						xxx SET @ xxx PSIG PSV 123 MINIMUM DISTANCE RELIEF SET POINTS SHALL BE INDICATED FOR ALL RELIEF LINES. ALL VAPOR SERVICE RELIEF VALVES SHALL BE INSTALLED OFF TOP VAPOR LINES. ALL RELIEF VALVE DISCHARGES SHALL ENTER THE TOP OF VENT/FLARE HEADERS.			16												
16	G GAUGING (DIMENSIONAL)			CC CORROSION COUPON			MWT MAXIMUM WORKING TEMPERATURE																					
	H HAND (MANUALLY)			CI CHEMICAL INJECTION			NC NORMALLY CLOSED									15												
	I CURRENT			CSC CAR SEAL CLOSE			MAWP MAXIMUM ALLOWABLE WORKING PRESSURE									14												
15	J SCAN (1) OR POWER (2)			CSO CAR SEAL OPEN			NCMH NORMAL CUBIC METER PER HOUR									13												
	K TIME			C&W COATED AND WRAPPED			NGL NATURAL GAS LIQUIDS																					
	L LEVEL			DE DE-ENERGIZED			NILL NORMAL INTERFACE LIQUID LEVEL																					
14	M MOISTURE/HUMIDITY			DP DESIGN PRESSURE			NNF NORMALLY NO FLOW																					
	N SHUTDOWN			ΔP DIFFERENTIAL PRESSURE			NO NORMALLY OPEN																					
	O UNCLASSIFIED (AS REQUIRED)			(F) FURNISHED BY VENDOR			NLL NORMAL LIQUID LEVEL																					
13	P PRESSURE/VACUUM			(E) EXISTING			OB OUTLET BASKET																					
	PD PRESSURE DIFFERENTIAL			E/E END TO END			OD OUTSIDE DIAMETER																					
	Q QUALITY (INTEGRATE OR TOTALIZE)			ES EMERGENCY STOP			OSBL OUTSIDE BATTERY LIMITS																					
12	R RADIOACTIVITY			EW EYE WASH			OWS OILY WATER SEWER																					
	S SPEED/FREQUENCY (1) OR SEAL (2)			EXT. EXTERNAL			PB PUSH BUTTON																					
	T TEMPERATURE			FC FAIL CLOSED			PSI POUNDS PER SQUARE INCH																					
11	TD TEMPERATURE DIFFERENTIAL			FL FAIL LAST			PSIG POUNDS PER SQUARE INCH GAUGE																					
	U MULTIVARIABLE			FMU FITTING MAKE-UP			RTD RESISTANCE TEMPERATURE DETECTOR																					
	V VISCOSITY (1) OR VIBRATION (2)			FO FAIL OPEN			RSD REACTOR SHUTDOWN																					
10	W WEIGHT/FORCE			FOB FLAT ON BOTTOM			S/FF SEAM TO FACE OF FLANGE																					
	WD WEIGHT DIFFERENTIAL			FOT FLAT ON TOP			SC SAMPLE CONNECTION																					
	X UNCLASSIFIED (AS REQUIRED)			FP FULL PORT VALVE			SEL SELECTOR SWITCH																					
	Y COMPUTING FUNCTION			FV FULL VACUUM			SO STEAM OUT																					
9	Z EMERGENCY ACTION (1) OR POSITION (2)			GC GAS CHROMATOGRAPH			SP SET POINT																					
				GO GEAR OPERATOR			STS SAFETY SHOWER																					
				HC HOSE CONNECTION			SS STAINLESS STEEL																					
				HH HAND HOLE			S/S SEAM TO SEAM																					
				HOA HAND-OFF-AUTO SELECTOR SWITCH			T/T TANGENT TO TANGENT																					
				HP HORSE POWER			T/FF TANGENT TO FACE OF FLANGE																					
				H.PT. HIGH POINT			TEG TRIETHYLENE GLYCOL																					
				HS HAND SWITCH			TSO TIGHT SHUTOFF																					
				HV HALF VACUUM			UG UNDERGROUND																					
				IA INSTRUMENT AIR SUPPLY			UL UNIT LIMITS																					
				IB INTERMITTANT BLOWDOWN			UC UTILITY CONNECTION																					
				I.D. INSIDE DIAMETER			US UTILITY STATION																					
				IP INTERMEDIATE PRESSURE			VB VORTEX BREAKER																					
5	LOCAL INSTRUMENT TAG			INSTRUMENT LOOP IDENTIFICATION			INSTRUMENT SYMBOLS (BUBBLES)										5											
4	XX - XXX			XXXX - XXX X XXX X X			LIGHT			FIELD MOUNTED INSTRUMENT			CONTROL ROOM PANEL MOUNTED INSTRUMENT (FRONT)			(DCS) DISTRIBUTED CONTROL SYSTEM (ACCESSIBLE TO OPERATOR)			THREADED INSTRUMENT CONNECTION TO PROCESS LINE AND EQUIPMENT			FLANGED INSTRUMENT CONNECTION TO PROCESS LINE AND EQUIPMENT			4			
	NUMBER IN SEQUENCE (BEGIN EACH FUNCTION WITH NUMBER 001)			SUFFIX NUMBER (IF REQUIRED)			FLOW COMPUTER (BY CLIENT)			LOCAL PANEL MOUNTED INSTRUMENT (FRONT)			CONTROL ROOM PANEL MOUNTED INSTRUMENT (REAR)			(DCS) DISTRIBUTED CONTROL SYSTEM (INACCESSIBLE TO OPERATOR)												
	FUNCTION LETTERS			SUFFIX LETTER (IF REQUIRED)																								
3				NUMBER IN SEQUENCE (MIN. 2 DIGITS)																								
				SUFFIX LETTER (IF REQUIRED)																								
				P&ID NUMBER (MIN. 2 DIGITS)																								
2				ISA FUNCTION LETTERS			LOCAL INTERLOCK			LOCAL PANEL MOUNTED INSTRUMENT (REAR)			INSTRUMENT WITH HEAT TRACING ET = ELECTRICAL TRACING ST = STEAM TRACING			ESD, BMS, PLC, PCS, SCS AND RTU INTERLOCK (ACCESSIBLE TO OPERATOR)						SOCKET WELDED INSTRUMENT CONNECTION TO PROCESS LINE AND EQUIPMENT			GENERIC INSTRUMENT CONNECTION TO PROCESS LINE AND EQUIPMENT			2
1																ESD, BMS, PLC, PCS, SCS AND RTU INTERLOCK (INACCESSIBLE TO OPERATOR)									1			
A							I INTERLOCKING FUNCTION W/IN EMERGENCY SHUTDOWN P PACKAGE INTERLOCK LOGIC																		B			
5677 Gauthier Road, New Franken, WI, 54229 agraenergy.com				DESIGNED BY:	NJS	--	XX-XX-XXXX	-----	---	---	DISTILLATION PIPING AND INSTRUMENTATION DIAGRAM				PROJECT:	NEW FRANKEN 1												
				DRAWN BY:	NJS	--	XX-XX-XXXX	-----	---	---					DRAWING #:	GENERAL												
				CHECKED BY:	---	01	07-22-2025	CONSTRUCTION ISSUE	NJS	---					SHEET #:	002												
				APPROVED BY:	---	NO.	DATE	REVISION	BY	APVD					DATE:	07-22-2025												

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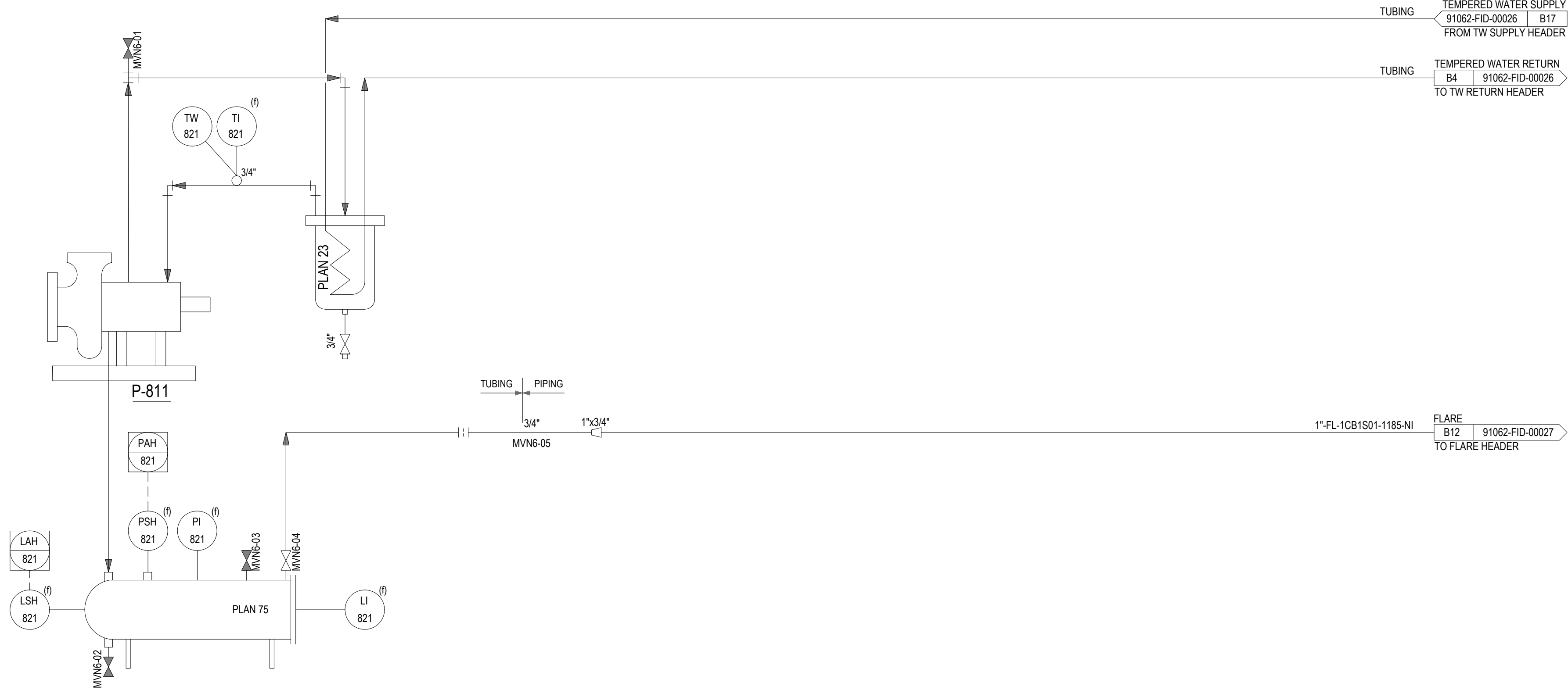
 5677 Gauthier Road, New Franken, WI, 54229 agraenergy.com	DESIGNED BY:	NJS	--	XX-XX-XXXX	-----	---	---	DISTILLATION PIPING AND INSTRUMENTATION DIAGRAM	PROJECT:	NEW FRANKEN 1
	DRAWN BY:	NJS	--	XX-XX-XXXX	-----	---	---		DRAWING #:	GENERAL
	CHECKED BY:	---	01	07-22-2025	CONSTRUCTION ISSUE	NJS	---		SHEET #:	003
	APPROVED BY:	---	NO.	DATE	REVISION	BY	APVD		DATE:	07-22-2025



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	DRAWN BY:	NJS	--	XX-XX-XXXX	-----	---		DRAWING #:	DISTILLATION
	CHECKED BY:	---	01	07-22-2025	CONSTRUCTION ISSUE	NJS		SHEET #:	017
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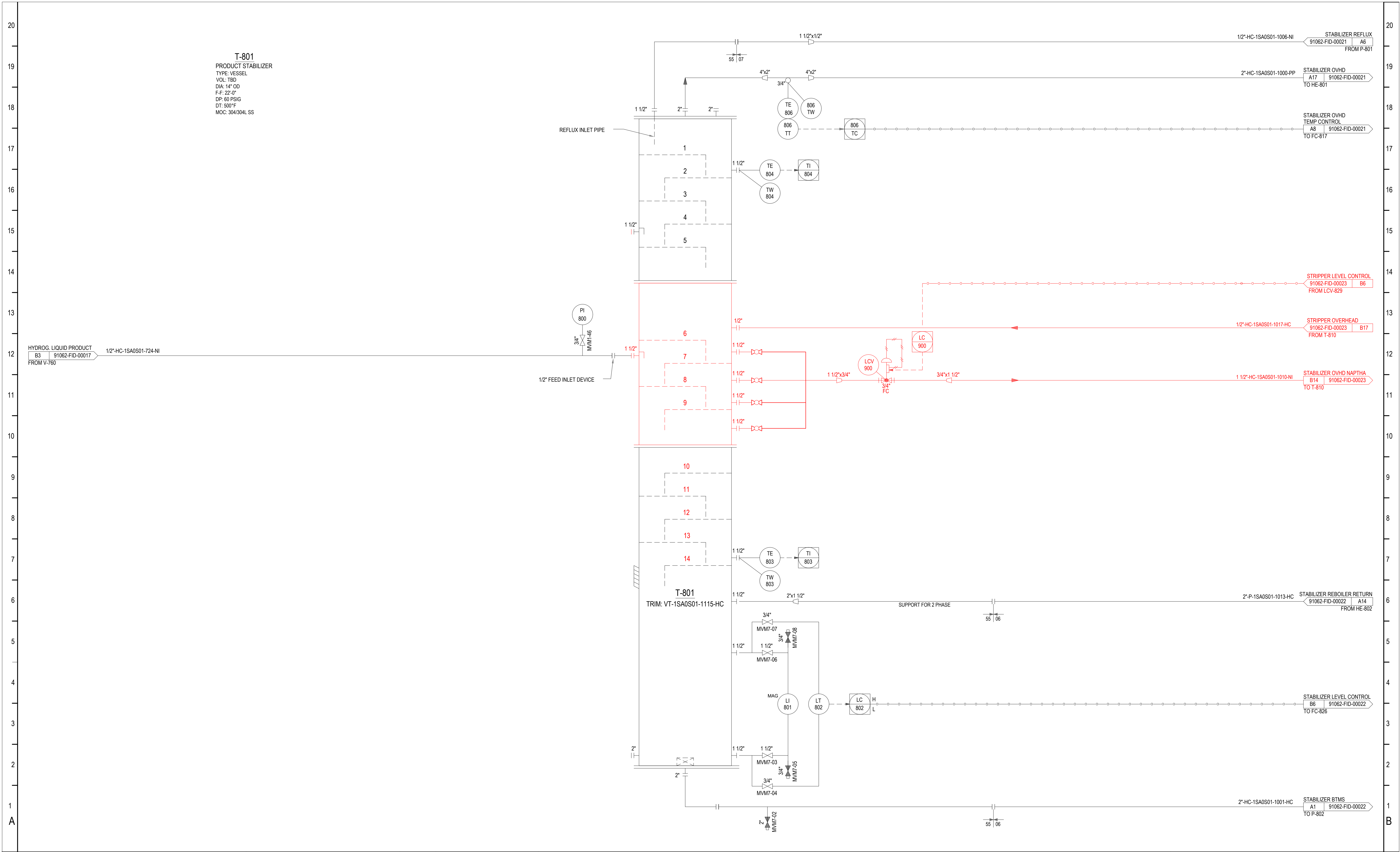
NOTES
1. COOLERS MAY BE INSTALLED IN HORIZONTAL OR VERTICAL HIGH POINT VENTS AND LOW POINT DRAINS ON BOTH PUMPAGE AND COOLANT CIRCUITS TO BE PROVIDED.



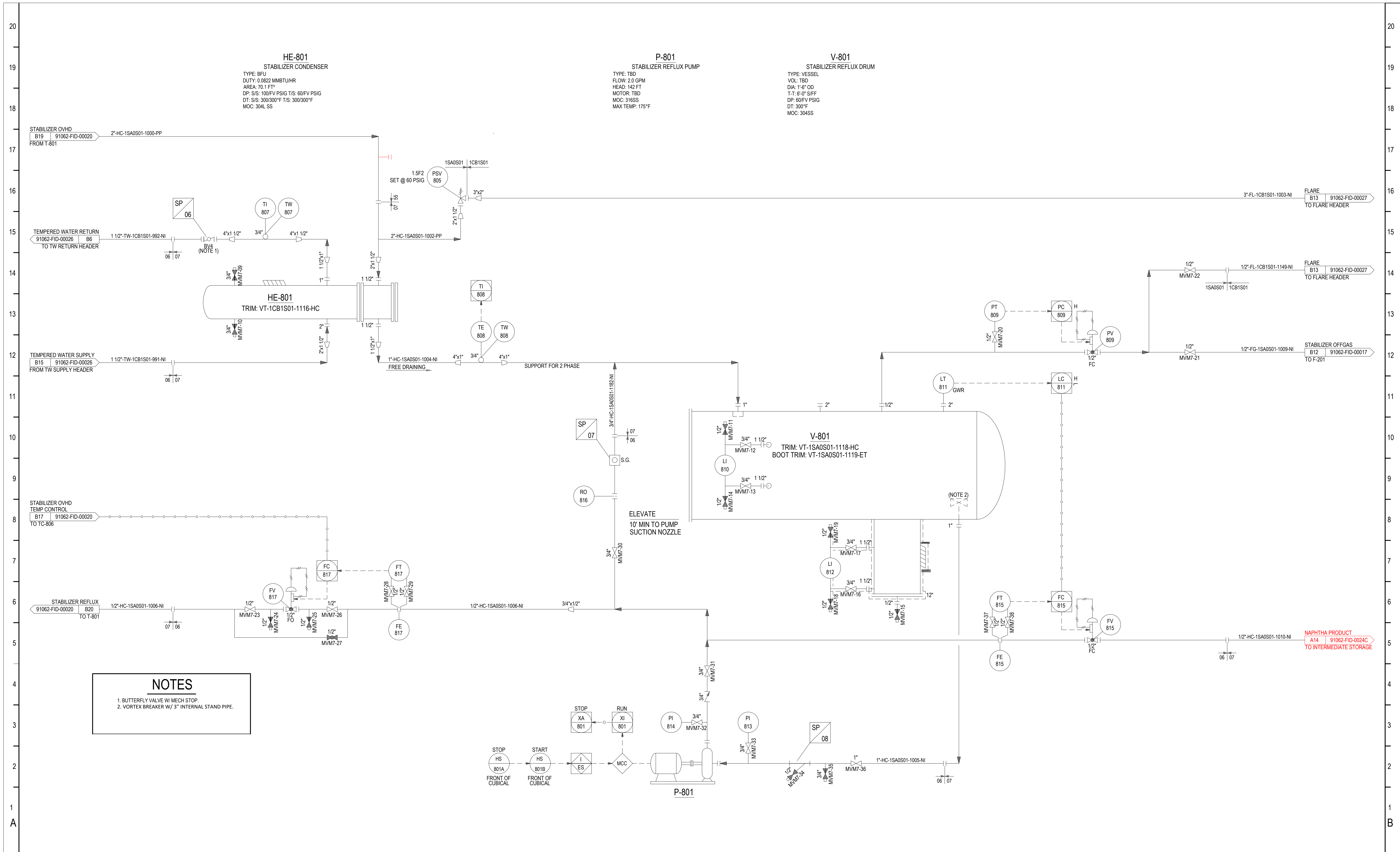
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 5677 Gauthier Road, New Franken, WI, 54229 agraenergy.com	DESIGNED BY:	NJS	--	XX-XX-XXXX	-----	---	---	PUMP SEAL PLAN 23 & 75 PIPIING AND INSTRUMENTATION DIAGRAM	PROJECT:	NEW FRANKEN 1
	DRAWN BY:	NJS	--	XX-XX-XXXX	-----	---	---		DRAWING #:	DISTILLATION
	CHECKED BY:	---	--	07-22-2025	CONSTRUCTION ISSUE	NJS	---		SHEET #:	019
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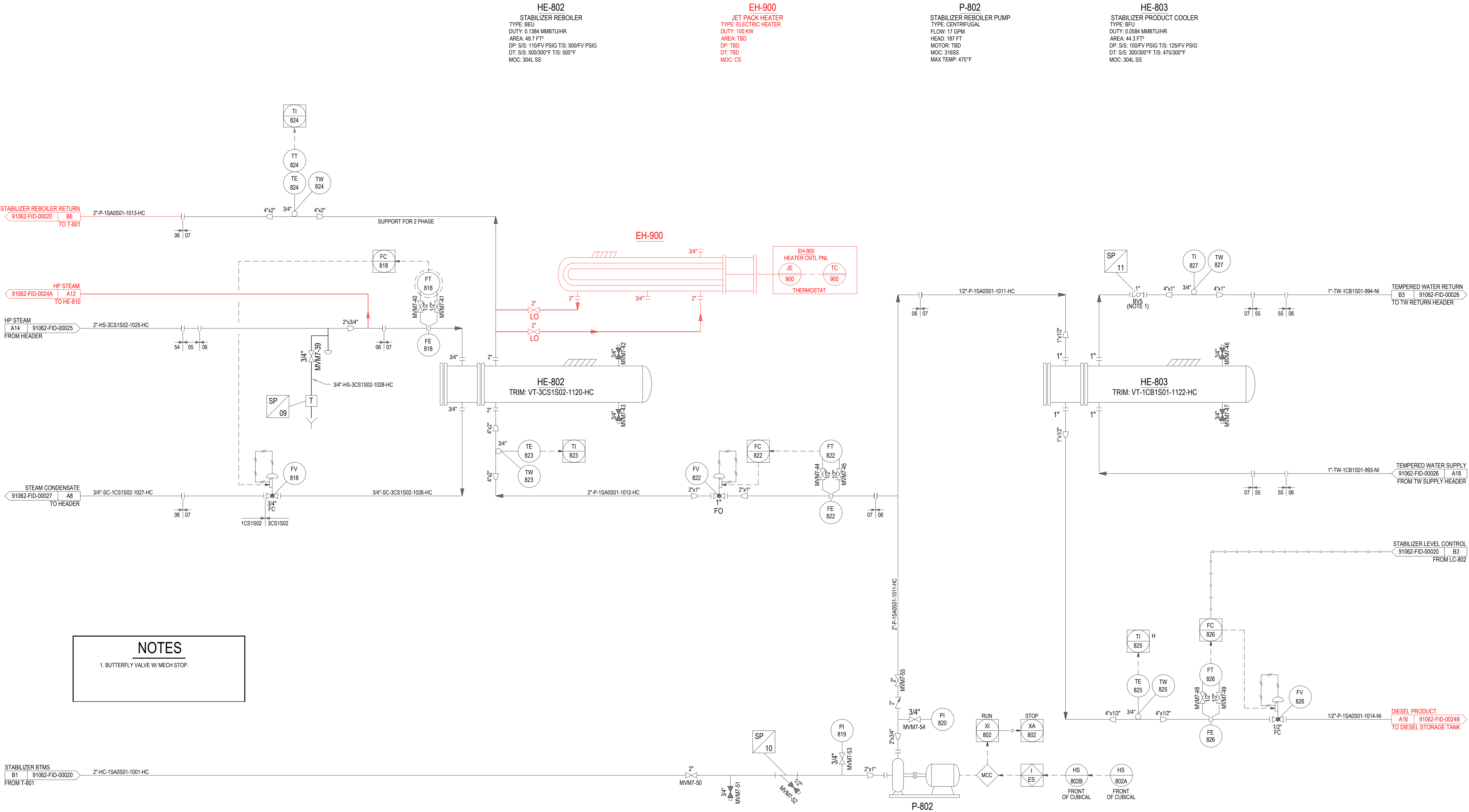


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STABILIZER BTMS REBOILER
AND PRODUCT PIPING AND
INSTRUMENTATION DIAGRAM

PROJECT:	NEW FRANKEN 1
DRAWING #:	DISTILLATION
SHEET #:	022
DATE:	07-22-2025



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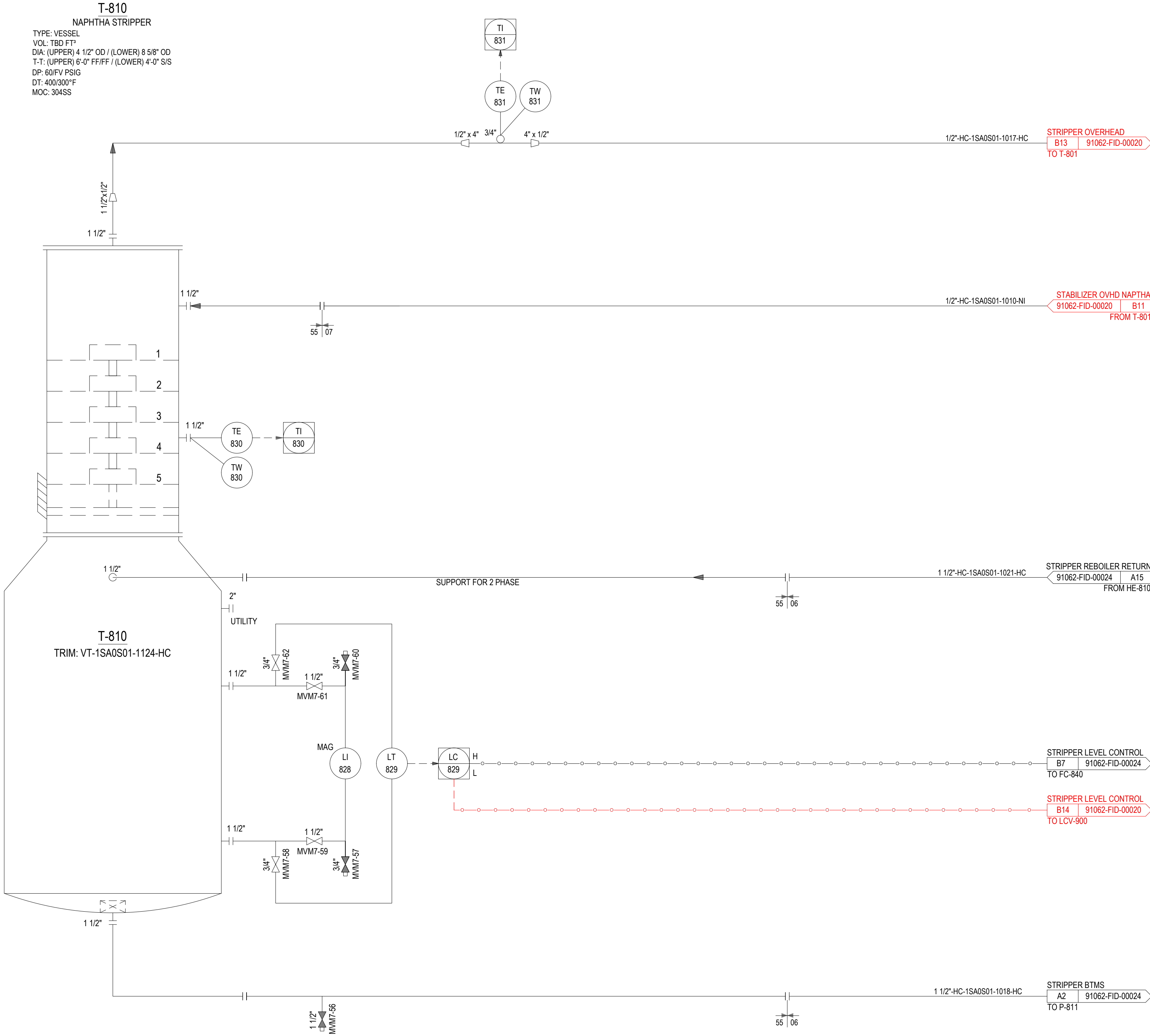


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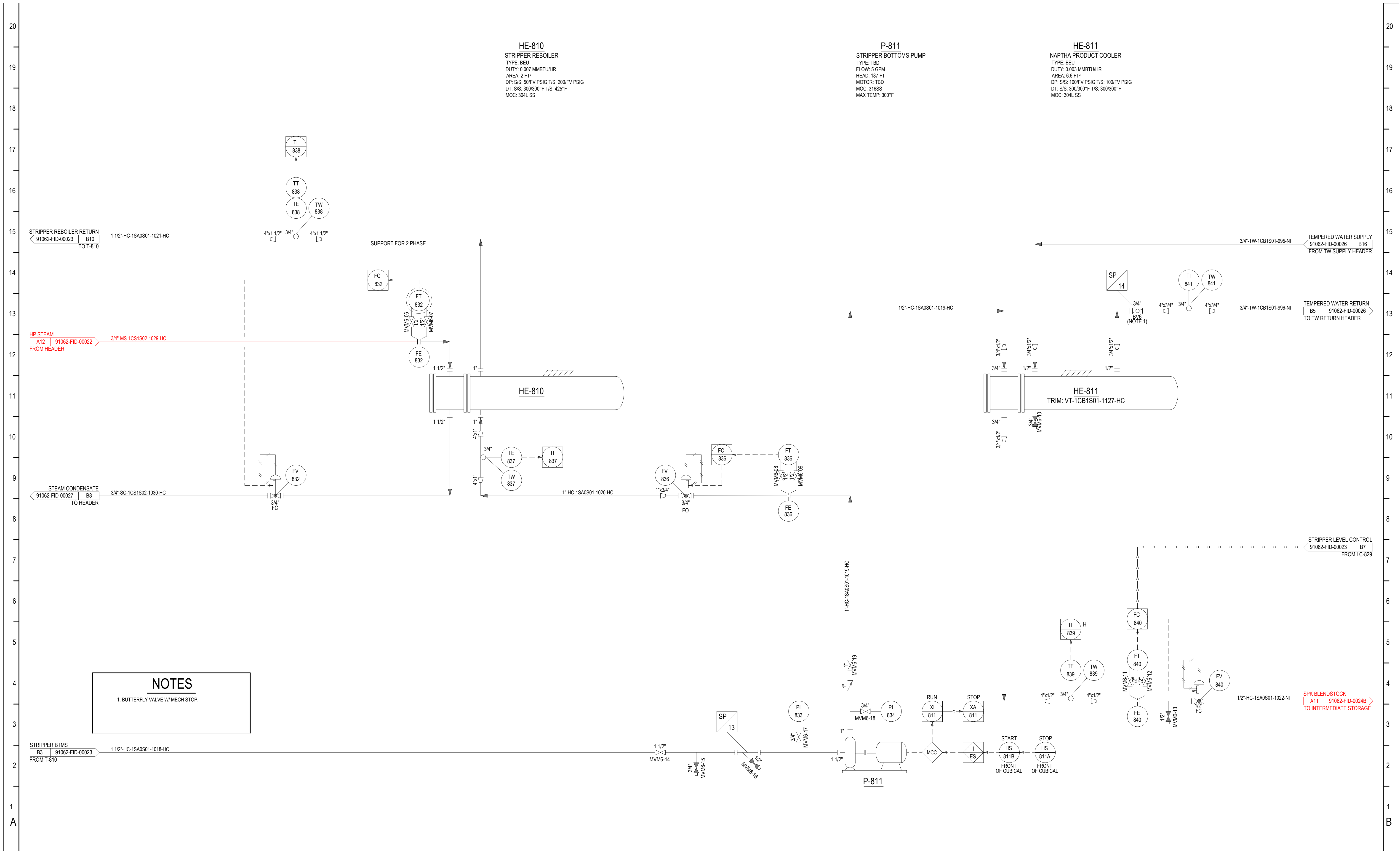
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APPROVED BY:	---	NO.	DATE	REVISION	BY	APVD

NAPHTHA STRIPPER PIPING
AND INSTRUMENTATION
DIAGRAM

PROJECT:	NEW FRANKEN 1
DRAWING #:	DISTILLATION
SHEET #:	023
DATE:	07-22-2025



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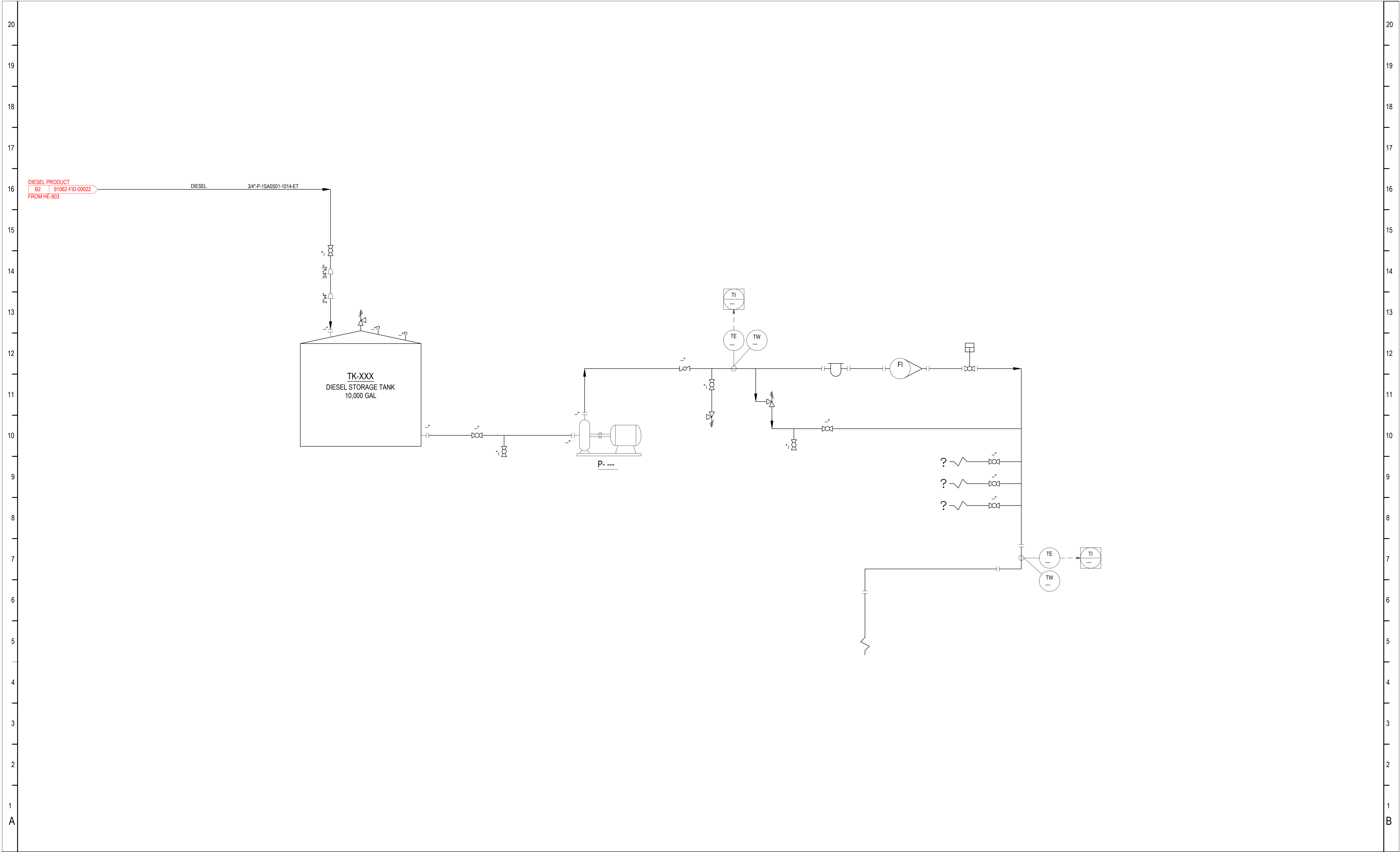
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APPROVED BY:	---	NO.	DATE	REVISION	BY	APVD

STRIPPER BTMS REBOILER & PRODUCT PIPING AND INSTRUMENTATION DIAGRAM

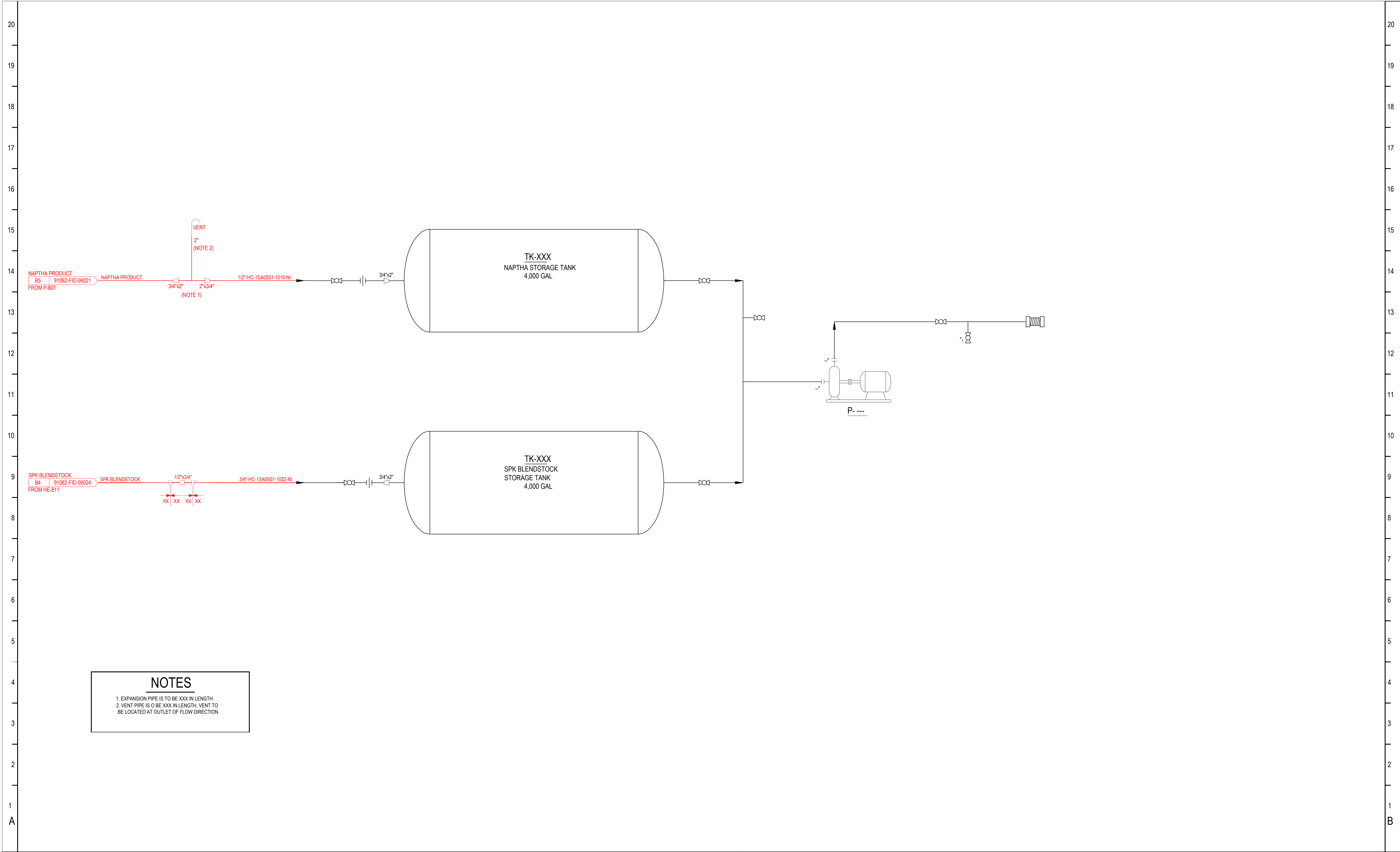
PROJECT:	NEW FRANKEN 1
DRAWING #:	DISTILLATION
SHEET #:	24A
DATE:	07-22-2025

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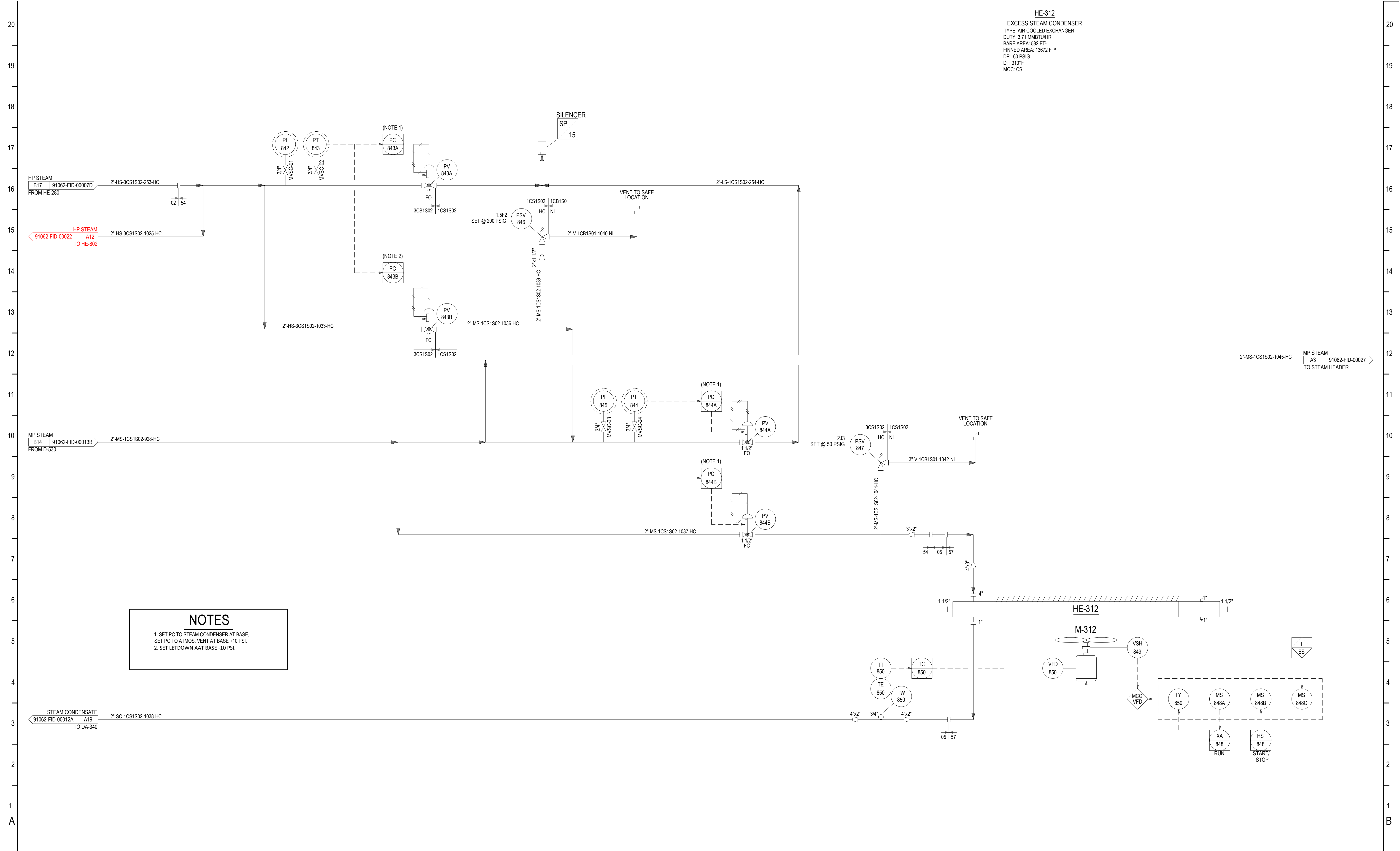
 5677 Gauthier Road, New Franken, WI, 54229 agraenergy.com	DESIGNED BY:	NJS	--	XX-XX-XXXX	-----	---	---	FUEL DEPOT PIPING AND INSTRUMENTATION DIAGRAM	PROJECT:	NEW FRANKEN 1
	DRAWN BY:	NJS	--	XX-XX-XXXX	-----	---	---		DRAWING #:	DISTILLATION
	CHECKED BY:	---	--	07-22-2025	CONSTRUCTION ISSUE	NJS	---		SHEET #:	24B
	APPROVED BY:	---	NO.	DATE	REVISION	BY	APVD		DATE:	07-22-2025

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	DRAWN BY:	NJS	--	XX-XX-XXXX	-----	---	---		DRAWING #:	STEAM	
	CHECKED BY:	---	--	07-22-2025	CONSTRUCTION ISSUE	NJS	---		SHEET #:	025	
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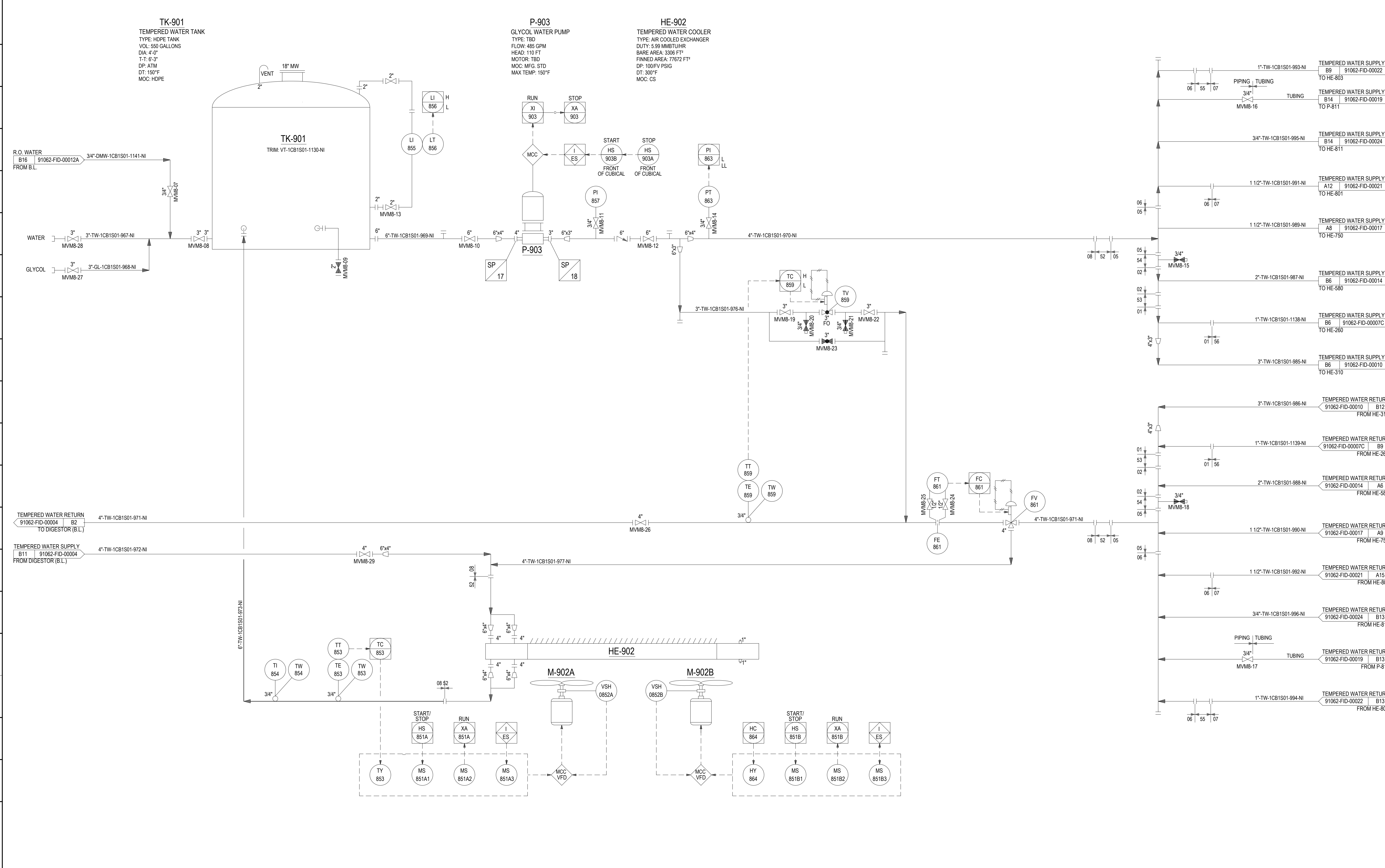


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APPROVED BY:	---	NO.	DATE	REVISION	BY	APVD

TEMPERED WATER SYSTEM
PIPIING AND
INSTRUMENTATION DIAGRAM

PROJECT:	NEW FRANKEN 1
DRAWING #:	TEMPERED WATER
SHEET #:	026
DATE:	07-22-2025



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APPROVED BY:	---	NO.	DATE	REVISION	BY	APVD

UTILITIES DISTRIBUTION
HEADERS PIPING AND
INSTRUMENTATION DIAGRAM

PROJECT:	NEW FRANKEN 1
DRAWING #:	UTILITIES
SHEET #:	027
DATE:	07-22-2025

